DS605: Fundamentals of Machine Learning

Lab Assignment - 4

End-to-End Machine Learning Project: Airbnb Price Prediction

Dataset: Kaggle New York City Airbnb Open Data (AB_NYC_2019.csv)
Submission: Public GitHub repository link and deployed application link, if available.

Objective

Build a complete machine learning workflow for Airbnb price prediction and make the final model usable through a simple web application.

End-to-End Machine Learning Workflow

Task 1 - Data Analysis and Preparation

- Analyze the Airbnb dataset and perform suitable cleaning, preprocessing, transformation, feature engineering, and feature selection. Handle missing values and outliers appropriately.
- Summarize the important data patterns, preprocessing decisions, and features that may influence Airbnb prices.

Task 2 - Model Training and Evaluation

- Train and compare suitable regression models, select the best model, perform appropriate hyperparameter tuning, and evaluate the final model using suitable regression metrics. Check for possible overfitting or underfitting and address it where required.
- Save the final preprocessing workflow and trained model so that the same pipeline can be used for new inputs.

Task 3 - Streamlit or Gradio Application

- Build a simple Streamlit or Gradio interface that accepts relevant Airbnb listing information and returns the estimated nightly price.
- Test the application with realistic inputs and deploy it online if possible.

Task 4 - Final Project Summary

- Present the main analysis, model comparison, final performance, application results, and important limitations of the developed system.

Deliverables (GitHub Only)

- Public GitHub repository containing the complete Jupyter Notebook, Streamlit or Gradio application, saved model/pipeline, requirements.txt, and README.
- Include model results, important plots, application screenshots, and the deployed application link if available.